DDP (stage 0)

all state replicated

observed per step:

5 all-reduces

ZeRO-2

gradients, optimizer

state partitioned;

observed: 13 all-

reduces, 1 all-gather

ZeRO-3

parameters also

partitioned; observed:

206 all-gathers,

11 reduce-scatters

Intel Max 1550

8 tiles per node

oneCCL over TCP,

host-staged

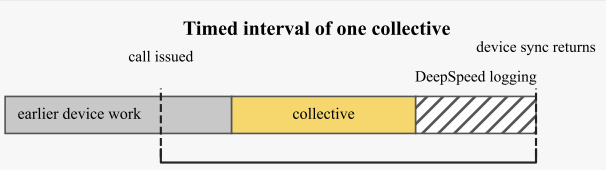
NVIDIA H100

4 GPUs per node; NVLink

within a node; NCCL over

InfiniBand, host-staged;

TCP-socket control



(a) state, observed collectives, and timer boundaries

Routing microbenchmark (two nodes)

one SwiGLU expert per rank, BF16

dispatch all-to-all → expert forward and backward

→ return all-to-all → final barrier wait

balanced vs. half of every rank's tokens to expert 0

(b) configurations and routing phases